

PART A : MCQs (25 questions. Right answer: +3, Wrong answer: -1)

- 1) Let $G(V,E)$ be a simple graph. Let $G'(V,E')$ be a graph obtained from G such that (u,v) is an edge in G' if (u,v) is not an edge in G . Which of the following is true?
 - (a) At least one of G or G' are connected.
 - (b) G is necessarily disconnected.
 - (c) Both G and G' are disconnected.
 - (d) None of the above.
- 2) For the inter-hostel six-a-side football tournament, a team of 6 players is to be chosen from 11 players consisting of 5 forwards, 4 defenders and 2 goalkeepers. The team must include at least 2 forwards, at least 2 defenders and at least 1 goalkeeper. Find the number of different ways in which the team can be chosen. (CMI 2014)
 - a. 260
 - b. 340
 - c. 720
 - d. 440
- 3) Find the coefficient of x^3 in $((1+x)^3)((2+x^2)^{10})$
 - (a) 2^{10}
 - (b) 2^{14}
 - (c) 2^9
 - (d) (I don't remember)
- 4) There are four machines and it is known that exactly two of them are faulty. They are tested one by one in a random order till both the faulty machines are identified. The probability that only two tests are required is:
 - (a) $1/2$
 - (b) $1/3$
 - (c) $1/4$
 - (d) $1/6$
- 5) Two fair dice are rolled. If the number on both the dice are divisible by 2, then your friend gives you Rs. 10, else you give your friend Rs. 3. The person who has more money wins the game. Who is expected to win?
 - (a) You.
 - (b) Your friend.
 - (c) Both.
 - (d) None.
- 6) For two n -bit strings $x, y \in \{0, 1\}^n$, define $z := x \oplus y$ to be the bitwise XOR of the two strings (that is, if x_i, y_i, z_i denote the i -th bits of x, y, z respectively, then $z_i = x_i + y_i \bmod 2$). A function $h : \{0, 1\}^n \rightarrow \{0, 1\}^n$ is called linear if $h(x \oplus y) = h(x) \oplus h(y)$, for every $x, y \in \{0, 1\}^n$. The number of such linear functions for $n \geq 2$:
 - (a) 2^n
 - (b) 2^{2n}

- (c) $2^{(n+1)}$
- (d) n

7) Which of these functions grow fastest with n ? (TIFR 2014)

- (a) $(e^n)/n$
- (b) $e^{(n-0.9\log n)}$
- (c) 2^n
- (d) $(\log n)^{(n-1)}$
- (e) None of the above.

8) A unit circle has ' n ' points on it such that the arcs made by these points are equal in size. Now, draw edges between every pair of these ' n ' points. What is the product of lengths of these chords?

- (a) 1
- (b) 2^n
- (c) $2^{(n*(n-1)/2)}$
- (d) $n^{(n/2)}$

9) Suppose you are given an array A with $2n$ numbers.

The numbers in odd positions are sorted in ascending order, that is, $A[1] \leq A[3] \leq \dots \leq A[2n-1]$.

The numbers in even positions are sorted in descending order, that is, $A[2] \geq A[4] \geq \dots \geq A[2n]$.

What is the method you would recommend for determining if a given number is in the array? (TIFR 2010)

- a. Sort the array using quick-sort and then use binary search.
- b. Merge the sorted lists and perform binary search.
- c. Perform a single binary search on the entire array.
- d. Perform separate binary searches on the odd positions and the even positions.
- e. Search sequentially from the end of the array.

10) Let $a_1, a_2, a_3, \dots, a_n$ be positive numbers. Then, the value must be

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1}$$

- (a) $\geq n$
- (b) $\leq n$
- (c) $\geq \sqrt{n}$
- (d) None.

11) For how many positive integers n is n^2+96 is a perfect square?

- a. 1
- b. 2
- c. 3
- d. 4

12) For each positive integer n consider the set S_n defined as follows: $S_1=\{1\}, S_2=\{2,3\}, S_3=\{4,5,6\}, \dots$ and in general, S_{n+1} consists of $n+1$ consecutive integers the smallest of which is one more than the largest integer in S_n . Then the sum of all the integers in S_{21} equals to (ISI 2017)

(a) 1113

(b) 53361

(c) 5082

(d) 4641

13) For $n \times n$ matrix such that all the principal diagonal elements equal to 2 and all other values being 1 what is the determinant of the matrix?

- a. $n+1$
- b. $2^n - 1$
- c. $n-1$
- d. $2^n + 1$

14) Consider the majority function on three bits, $\text{maj}: \{0,1\}^3 \rightarrow \{0,1\}$ where $\text{maj}(x_1, x_2, x_3) = 1$ if and only if $x_1 + x_2 + x_3 \geq 2$. Let $p(\alpha)$ be the probability that the output is 1 when each input is set to 1 independently with probability α . What is $p'(\alpha) = d/d\alpha p(\alpha)$? (TIFR 2017)

- a. 3α
- b. α^2
- c. $6\alpha(1-\alpha)$
- d. $3\alpha^2(1-\alpha)$

15) Let r, s, t be regular expressions. Which of the following identities is correct? (TIFR 2010)

- a. $(r+s)^* = r^*s^*$
- b. $r(s+t) = rs+rt$
- c. $(r+s)^* = r^*+s^*$
- d. $(rs+r)^*r = r(sr+r)^*$
- e. $(r^*s)^* = (rs)^*$

16) Let $G=(V,E)$ be a simple undirected graph on n vertices. A colouring of G is an assignment of colours to each vertex such that endpoints of every edge are given different colours. Let $\chi(G)$ denote the chromatic number of G , i.e. the minimum numbers of colours needed for a valid colouring of G . A set $B \subseteq V$ is an independent set if no pair of vertices in B is connected by an edge. Let $\alpha(G)$ be the number of vertices in a largest possible independent set in G . In the absence of any further information about G we can conclude. (TIFR 2013)

- a. $\chi(G) \geq \alpha(G)$
- b. $\chi(G) \leq \alpha(G)$

- c. $a(G) \geq n\chi(G)$
- d. $a(G) \leq n\chi(G)$
- e. None of the above.

17) Consider the following recurrence relation: (TIFR 2014)

$$T(n) = T(n/k) + T(3n/4) + n \text{ if } n \geq 2$$

$$1 \text{ if } n=1$$

Which of the statements is FALSE?

- a. $T(n)$ is $O(n^{3/2})$ when $k=3$.
- b. $T(n)$ is $O(n \log n)$ when $k=3$.
- c. $T(n)$ is $O(n \log n)$ when $k=4$.
- d. $T(n)$ is $O(n \log n)$ when $k=5$.
- e. $T(n)$ is $O(n)$ when $k=5$.

8. Let A and B be finite sets such that $A \subseteq B$. Then, what is the value of the expression:

$$\sum_{C: A \subseteq C \subseteq B} (-1)^{|C \setminus A|},$$

where $C \setminus A = \{x \in C : x \notin A\}$?

18)

a.

Always 0

b. Always 1

c. 0 if $A=B$ and 1 otherwise

d. 1 if $A=B$ and 0 otherwise

e. Depends on the size of the universe

19) How many pairs of sets (A, B) are there that satisfy the condition A, B subset $\{1, 2, 3, 4, 5\}$, $A \cap B = \emptyset$? (Ans 243)

- a. 125
- b. 127
- c. 130
- d. 243
- e. 257

20) What is the minimum number of students needed in a class to guarantee that there are at least 6 students whose birthdays fall in the same month ?

- a. 6
- b. 23
- c. 61
- d. 72
- e. 91

21) which of the following is true for a Nilpotent matrix?

- a. Zero is one of the eigen value.
- b. Determinant is always zero
- c. Determinant is always non zero
- d. It is a non singular matrix.

22) Find the largest positive value of n for which $n^3 + 100$ is divisible by $n+10$.

- a. 890
- b. 90
- c. 1390
- d. (don't remember)

23) $L = \{x \mid x \in (0+1)^* \text{ length of } x \text{ is divisible by } 3\}$ Number of states required in minimized DFA for the language?

- a. 3
- b. 4
- c. 5
- d. 2

24) A unit circle on which points are uniformly distributed such that each point makes arc of equal length, now construct chord between every pair of points on the circle, find the product of the length of the chords.

- a. $2^{(nC_2)}$
- b. Don't remember
- c. Don't remember
- d. Don't remember

25) In graph G , where r is a source vertex in BFS traversal vertex u is visited before vertex v then which of the following is always true?

- a. $d(r, u) < d(r, v)$
- b. $d(r, u) \leq d(r, v)$

- c. $d(r, u) > d(r, v)$
- d. $d(r, u) \geq d(r, v)$

PART B (There were 5 questions of 8 marks out of which 3 had to be attempted and 4 questions of 12 marks out of which 2 had to be attempted)

8 MARK QUESTIONS:

- 1) Let X and Y be two sets and ' f ' be a function such that $f: X \rightarrow Y$. Show that the following 2 statements are equivalent.
 - (a) f is one to one.
 - (b) For all sets ' Z ' such that $g_1: Z \rightarrow X$ and $g_2: Z \rightarrow X$ and $g_1 \neq g_2$ implies $f(g_1) \neq f(g_2)$
- 2) A *trail* is a walk such that no edge is repeated. Consider you have an ' n ' vertex complete graph, with each edge labeled $1, 2, 3, \dots, n*(n-1)/2$. Show that there exists a trail of length at least $(n-1)$ that is strictly increasing. (HINT : Assign a weight ' w ' to each vertex ' x ' whose value is the length of the longest increasing trail such that ' x ' is the last vertex of that trail. Now, show that $\sum_w w(x) = n*(n-1)$. Using this prove the above statement)
- 3) Let A be a finite state automaton having Q states, I initial states and F final states. A string is said to be in the language if $\delta(I, w)$ leads to one of the states in ' F '. Show that the language can also be accepted if :
 - (a) Number of initial states(I) = 2
 - (b) Number of final states(F) = 2
- 4) You have an incoming stream of numbers. You have to choose a number to represent the data you have seen so far such that the chosen number should be uniformly distributed over the numbers seen so far. That is, If you have seen $x_1, x_2, \dots, x_i$ so far, the item you pick to represent the data set must be one of these numbers. $\Pr[X=x_i] = 1/i$. You do not have space to store all the numbers and you cannot revisit any item again because it is a stream. Describe an algorithm to achieve this efficiently. (HINT: You assign a probability to decide whether to replace the item based on the new element that arrives)
- 5) A directed graph $G = (V, E)$ is said to be semi-connected if, for all pairs of vertices u, v in V we have $u \rightarrow v$ or $v \rightarrow u$ path.
 - (a) Give an example of DAG that is NOT semi connected.
 - (b) Describe and analyze an algorithm that efficiently determines if a given DAG is semi connected or not.
 - (c) Describe and analyze an algorithm that efficiently determines if an arbitrary graph (not necessarily acyclic) is semi connected or not.

12 MARK QUESTIONS:

- 1) Let ' S ' be a string of length ' n ' defined over alphabet Σ . A 3-cadence is defined as:
 $S[i] = S[i+d] = S[i+2d]$

And $n < (i+3d)$.

- (a) Give an $O(n^2 * \text{length of } (\Sigma))$ algorithm to find if a given string has a 3-cadence.
- (b) (I don't remember)

- 2) (a) Prove that if sum of degrees of every pair of non adjacent vertices = n , then the given graph is Hamiltonian.

(b) Prove that if degree of every vertex is $\geq (n/2)$, the given graph is Hamiltonian.

- 3) Let there be a sequence of length ' n '. If $n > s * p * r$, then show that there exists either a strictly increasing subsequence of length ' s ' or a strictly decreasing subsequence of length ' r ' or a constant subsequence of length ' p '.
- 4) Some problem related to Convolution..